## Lineer Zamanla Değişmeyen (LZD) Sistemler:

Linear Time-Invariant (LTI) Systems

LTI sistemler detaylı olarak analiz edilebilirler

- Üst üste ekleme (superposition) özelliği
- Dürtü cevabı (impulse response) ile tüm karakteristik özellikleri belirlenebilir.

**Dürtü cevabı**: Bir LTI sistemin dürtü cevabı $h(t)/h[n]$, sisteme birim dürtü işareti $\delta(t)/\delta[n]$ uygulandığında elde edilen çıkıştır.

$$x(t)=\delta(t) \longrightarrow y(t)=h(t)$$

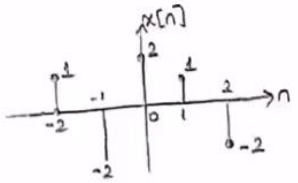
$$x[n]=\delta[n] \longrightarrow y[n]=h[n]$$

⇒ Herhangi bir $x[n]/x(t)$ işareti, zamanda kaydırılmış birim dürtü işaretlerinin lineer kombinasyonu ile ifade edilebilir.

$$x[n]=\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]=\cdots+x[-1]\,\delta[n+1]+x[0]\,\delta[0]+x[1]\,\delta[n-1]+\cdots$$

$$x(t)=\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau$$

Ex:

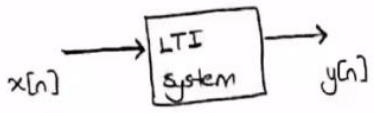



Birim dürtü işareti cinsinden ifade ediniz.

$$x[n]=\sum_{k=-2}^{2} x[k]\,\delta[n-k]=x[-2]\,\delta[n+2]+x[-1]\,\delta[n+1]+x[0]\,\delta[n]+x[1]\,\delta[n-1]+x[2]\,\delta[n-2]$$

$$=\delta[n+2]-2\,\delta[n+1]+2\,\delta[n]+\delta[n-1]-2\,\delta[n-2]$$

## Ayrık zamanlı LTI sistemlerde konvolüsyon




$x[n] = \delta[n] \longrightarrow y[n] = h[n]$ : Birim dürtü cevabı

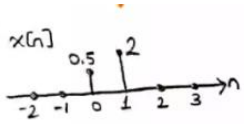
$x[n] = \delta[n-1] \longrightarrow y[n] = h[n-1]$

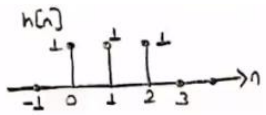
$x[n] = 3\,\delta[n+2] \longrightarrow y[n] = 3\,h[n+2]$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$ : Ayrık zamanlı konvolüsyon

$$x[n] \xrightarrow{\text{LTI } h[n]} y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

Ex: $x[n]$ ve $h[n]$ verilmiş, $y[n] = ?$



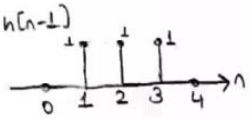




$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = x[0]h[n] + x[1]h[n-1] = 0.5\,h[n] + 2h[n-1]$$

$k \in (0,1)$

$n \in (0,1,2)$ $\quad$ $n \in (1,2,3)$

$n \in (0,1,2,3)$




$y[n] = 0, \quad n \leq -1$

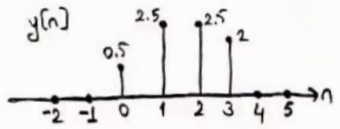
$y[0] = 0.5\,h[0] + 2\,h[-1] = 0.5$ ($h[-1] = 0$)

$y[1] = 0.5\,h[1] + 2h[0] = 2.5$

$y[2] = 0.5\,h[2] + 2h[1] = 2.5$

$y[3] = 0.5\,h[3] + 2h[2] = 2$ ($h[3] = 0$)

$y[n] = 0, \quad n \geq 4$

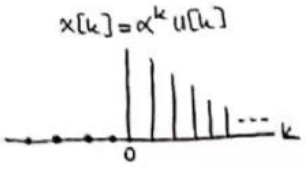


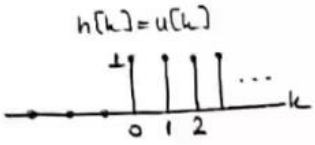
$0 < \alpha < 1$

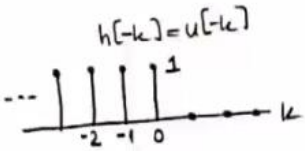
Ex: $x[n] = \alpha^n u[n]$ , $h[n] = u[n]$, $y[n] = ?$

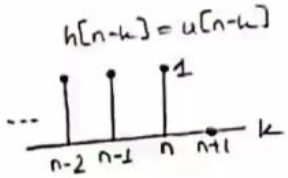
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} \alpha^k u[k]\,u[n-k] = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}\,u[n]$$

($k \geq 0$, $n-k \geq 0$, $n \geq k$)

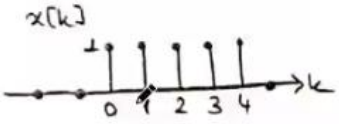



Ex: $x[n] = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{diğerde} \end{cases}$ $h[n] = \begin{cases} \alpha^n, & 0 \leq n \leq 6 \\ 0, & \text{diğerde} \end{cases}$ $y[n] = ?$ $\alpha > 1$







$n < 0$ $\quad y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = 0$

$0 \leq n \leq 4$ $\quad y[n] = \sum_{k=0}^{n} x[k]\,h[n-k] = \sum_{k=0}^{n} \alpha^{n-k} = \sum_{r=0}^{n} \alpha^r = \frac{1-\alpha^{n+1}}{1-\alpha}$ ($n-k=r$)

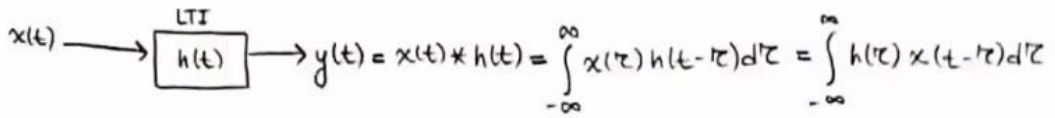
$\left.\begin{matrix} n > 4 \\ n-6 \leq 0 \end{matrix}\right\}$ $4 < n \leq 6$ $\quad y[n] = \sum_{k=0}^{4} \alpha^{n-k} = \alpha^n \sum_{k=0}^{4} \alpha^{-k} = \alpha^n \cdot \frac{1-(\alpha^{-1})^5}{1-(\alpha^{-1})} = \frac{\alpha^{n-4} - \alpha^{n+1}}{1-\alpha}$

$\left.\begin{matrix} n > 6 \\ n-6 \leq 4 \end{matrix}\right\}$ $6 < n \leq 10$ $\quad y[n] = \sum_{k=n-6}^{4} \alpha^{n-k} = \sum_{r=0}^{10-n} \alpha^{6-r} = \alpha^6 \sum_{r=0}^{10-n} \alpha^{-r} = \alpha^6 \frac{1-\alpha^{n-11}}{1-\alpha^{-1}} = \frac{\alpha^{n-4} - \alpha^7}{1-\alpha}$ ($r = k-n+6$)

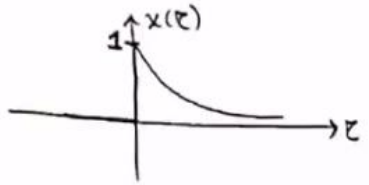
$n-6 > 4 \rightarrow n > 10$ $\quad y[n] = 0$

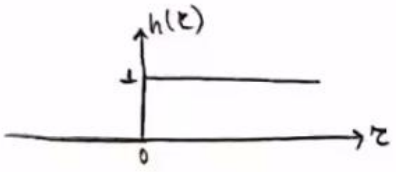
$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1-\alpha^{n+1}}{1-\alpha}, & 0 \leq n \leq 4 \\ \frac{\alpha^{n-4} - \alpha^{n+1}}{1-\alpha}, & 4 < n \leq 6 \\ \frac{\alpha^{n-4} - \alpha^7}{1-\alpha}, & 6 < n \leq 10 \\ 0, & n > 10 \end{cases}$$

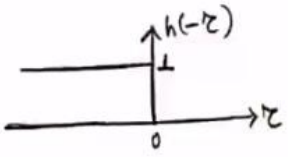
## Sürekli Zamanlı LTI Sistemlerde Konvolüsyon

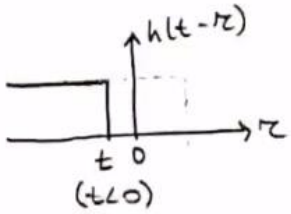



Ex: $x(t) = e^{-at}u(t)$, $a>0$ , $h(t)=u(t)$ , $y(t)=?$



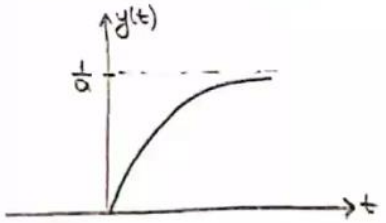




$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

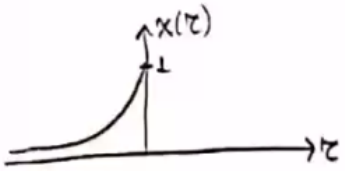
$t<0$ $\quad y(t)=0$

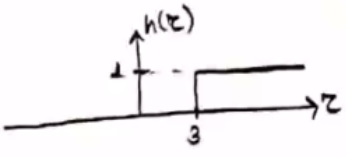
$t\geq 0$ $\quad y(t) = \int_0^t e^{-a\tau}d\tau = \frac{e^{-a\tau}}{-a}\Big|_0^t = \frac{1}{a}(1-e^{-at})$

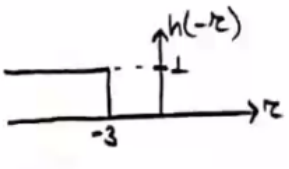
$$y(t) = \frac{1}{a}\left(1-e^{-at}\right)u(t)$$

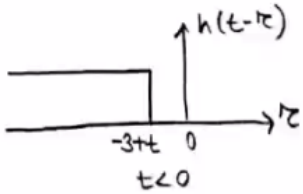



Ex: $x(t) = e^{2t}u(-t)$ , $h(t)=u(t-3)$ , $y(t)=?$



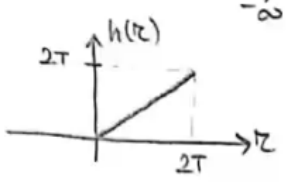
$t-3 \leq 0 \quad y(t) = \int_{-\infty}^{t-3} e^{2\tau} d\tau = \frac{1}{2} e^{2\tau} \Big|_{-\infty}^{t-3} = \frac{1}{2} e^{2(t-3)}$

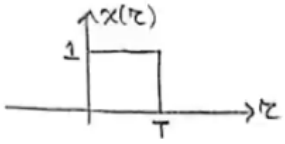
$t-3 > 0 \quad y(t) = \int_{-\infty}^{0} e^{2\tau} d\tau = \frac{1}{2}$

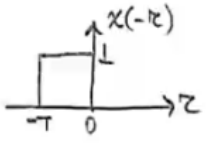
$$y(t) = \begin{cases} 0.5\, e^{2(t-3)}, & t \leq 3 \\ 0.5, & t > 3 \end{cases}$$

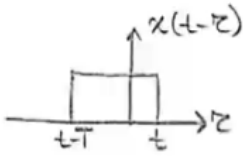
Ex: $x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{dışında} \end{cases}$ , $h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{dışında} \end{cases}$ , $y(t) = ?$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$




$t < 0 \quad y(t) = 0$

$0 < t < T \quad y(t) = \int_{0}^{t} \tau d\tau = \frac{\tau^2}{2} \Big|_{0}^{t} = t^2/2 \quad \leftarrow \begin{matrix} t > 0 \\ t-T < 0 \end{matrix}$

$T < t < 2T \quad y(t) = \int_{t-T}^{T} \tau d\tau = \frac{\tau^2}{2} \Big|_{t-T}^{T} = \frac{T^2}{2} - \frac{(t-T)^2}{2} = tT - T^2/2 \quad \leftarrow \begin{matrix} t < 2T \\ t-T > 0 \end{matrix}$

$2T < t < 3T \quad y(t) = \int_{t-T}^{2T} \tau d\tau = \frac{\tau^2}{2} \Big|_{t-T}^{2T} = -\frac{t^2}{2} + Tt + \frac{3T^2}{2} \quad \leftarrow \begin{matrix} t > 2T \\ t-T < 2T \end{matrix}$

$t > 3T \quad y(t) = 0$

$$y(t) = \begin{cases} 0, & t < 0 \\ t^2/2, & 0 < t < T \\ tT - T^2/2, & T < t < 2T \\ -t^2/2 + Tt + 3T^2/2, & 2T < t < 3T \\ 0, & t > 3T \end{cases}$$

## Konvolüsyon işleminin özellikleri

### 1) Yer değiştirme (commutative) özelliği

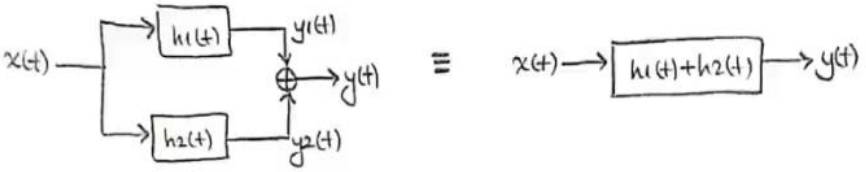
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$$

### 2) Dağılma (distributive) özelliği

$$x[n] * \big(h_1[n] + h_2[n]\big) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \big[h_1(t) + h_2(t)\big] = x(t) * h_1(t) + x(t) * h_2(t)$$

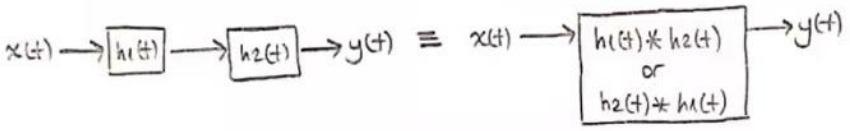



Dağılma özelliği ≡ Paralel Bağlama (only for LTI)

### 3) Birleşme (associative) özelliği

$$x[n] * \big(h_1[n] * h_2[n]\big) = \big(x[n] * h_1[n]\big) * h_2[n] = x[n] * h_1[n] * h_2[n]$$

$$x(t) * \big[h_1(t) * h_2(t)\big] = \big[x(t) * h_1(t)\big] * h_2(t) = x(t) * h_1(t) * h_2(t)$$




Birleşme özelliği ≡ Seri Bağlama (only for LTI)

## LTI sistemlerin özellikleri

LTI sistemlerin özellikleri birim dürtü cevabı üzerinden belirlenebilir.

### 1) Hafızaya sahip olan ve olmayan LTI sistemler (LTI systems with and without memory)

Bellekli ve bellekisiz LTI sistemler

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

Reminder: Bir sistemin herhangi bir zamandaki çıkışı, sadece o andaki girişe bağlı ise bellek sizdir.

Bellek siz: $h[n] = 0,\ n \neq 0$

$$h[n] = K\,\delta[n] \longrightarrow y[n] = x[n] * h[n] = K\,x[n] * \delta[n] = K\,x[n]$$

Reminder: $x[n] * \delta[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] = x[n]$

h[n]=1, n=0,1
=d[n]+d[n-1]

x[n]
y[n]=x[n]+x[n-1]

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

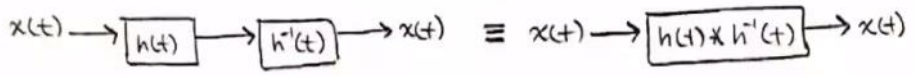
$h(t) = 0,\ t \neq 0$

$$h(t) = K\,\delta(t) \longrightarrow y(t) = K\,x(t)$$

Reminder: $x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau = x(t)$

## 2) Tersinir ve tersinir olmayan LTI sistemler:

Reminder: Bir sistem, tersi ile seri bağlandığında elde edilen çıktı ile girdi aynı ise sistem tersinirdir




$h(t) * h^{-1}(t) = \delta(t)$

Tersinir: $h(t) * h^{-1}(t) = \delta(t)$

⇒ LTI bir sistemin tersi de LTI'dir.

$$h[n] * h^{-1}[n] = \delta[n]$$

Ex: $y(t) = x(t-t_0)$ sistemi tersinir midir?

$$y(t) = x(t) * h(t) = x(t-t_0) \longleftarrow h(t) = \delta(t-t_0) \longrightarrow y(t) = x(t) * \delta(t-t_0) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-t_0-\tau)\,d\tau$$

$$= \int_{-\infty}^{\infty} x(t-t_0)\,\delta(t-t_0-\tau)\,d\tau$$

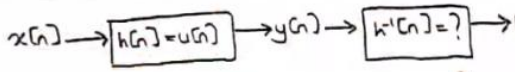
$$= x(t-t_0) \int_{-\infty}^{\infty} \delta(t-t_0-\tau)\,d\tau = x(t-t_0)$$

$h(t) * h^{-1}(t) = \delta(t)$

$h^{-1}(t) = \delta(t+t_0)$

$\delta(t-t_0) * \delta(t+t_0) = \delta(t)$ Tersinirdir.

Ex: Birim dürtü cevabı $h[n] = u[n]$ olan LTI bir sistem tersinir midir.




$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]u[n-k] = \sum_{k=-\infty}^{n} x[k]$$

$$w[n] = y[n] - y[n-1] = \sum_{k=-\infty}^{n} x[k] - \sum_{k=-\infty}^{n-1} x[k] = \left(\cdots + x[n-1] + x[n]\right) - \left(\cdots + x[n-2] + x[n-1]\right) = x[n]$$

$h^{-1}[n] = \delta[n] - \delta[n-1]$

$h[n] * h^{-1}[n] \stackrel{?}{=} \delta[n]$

$u[n] * \left(\delta[n] - \delta[n-1]\right) = u[n] * \delta[n] - u[n] * \delta[n-1] = u[n] - u[n-1] = \delta[n]$ ✓ Tersinir.